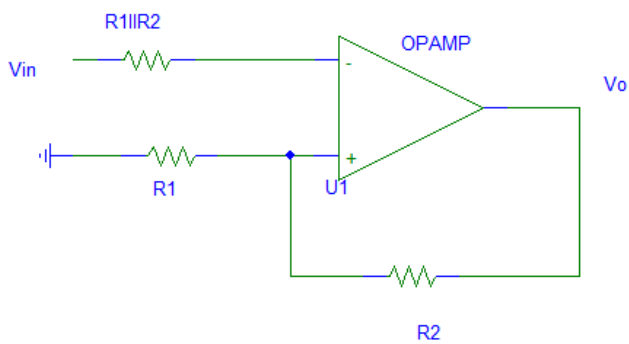
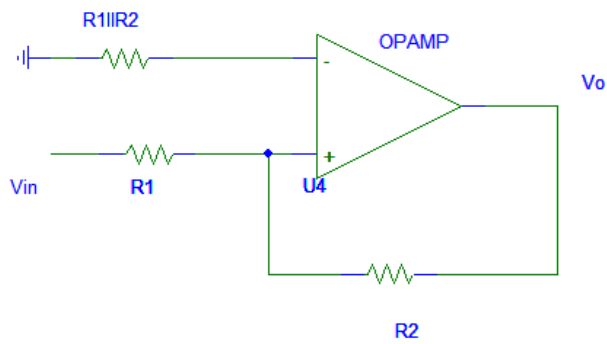


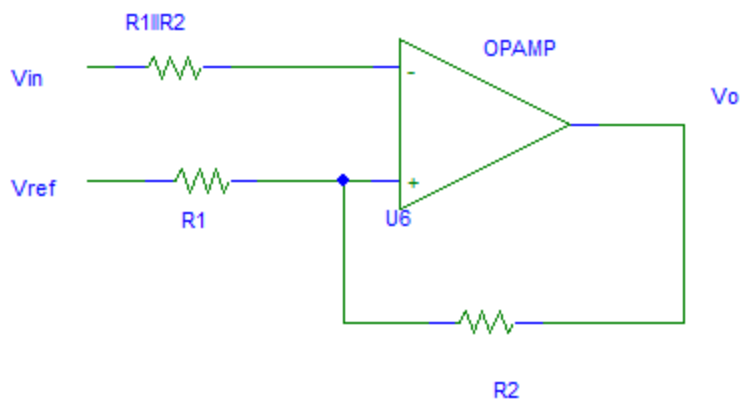
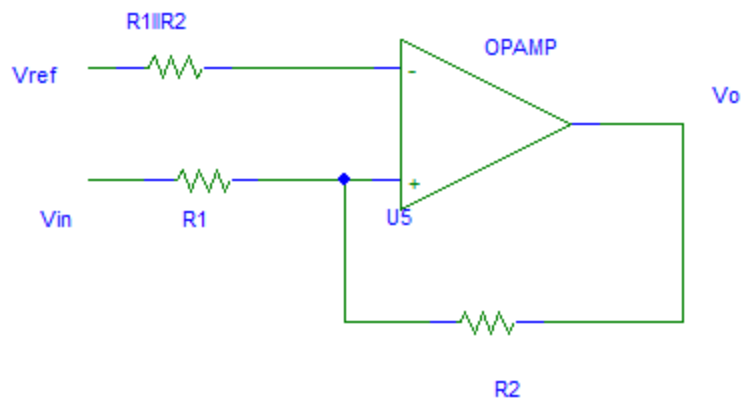
4.



Here, $R1 = 5k$, $R2 = 10k$, $V_{ref} = 2V$, $V_H = +10V$ and $V_L = -5V$

- Identify the type of the above schmitt trigger circuits and compare their input output characteristics.
- What will be their higher threshold voltage, lower threshold voltage and hysteresis width?
- Draw their transfer characteristics.
- Is it possible to shift their center voltage from the origin with little modification of the circuits? Draw new circuits with this modification.

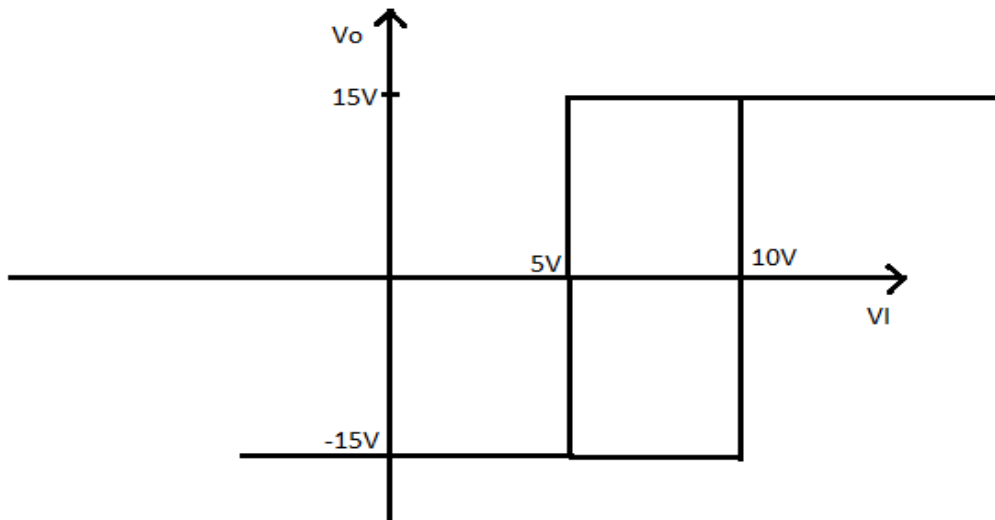
5.



Here, $R1 = 5k$, $R2 = 10k$, $V_{ref} = 2V$, $V_H = +10V$ and $V_L = -10V$

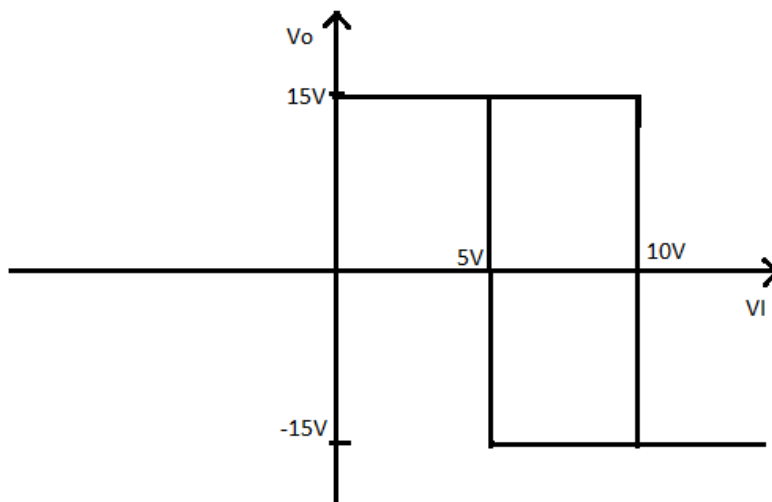
- Identify the type of the above schmitt trigger circuits and compare their input output characteristics.
- What will be their higher threshold voltage, lower threshold voltage, shift voltage and hysteresis width?
- Draw the voltage transfer characteristics curve (V_{in} vs V_{out} plot). Clearly label the plot.

6.



- Identify the schmitt circuit from the VTC.
- What is the hysteresis width, Shift voltage, V_H , V_L , V_{TH} and V_{TL} from this VTC.
- Design a schmitt trigger that will produce the same VTC.

7.



- Identify the schmitt circuit from the VTC.
- What is the hysteresis width, Shift voltage, V_H , V_L , V_{TH} and V_{TL} from this VTC.
- Design a schmitt trigger that will produce the same VTC.

*8. Suppose you want to design a street light controller. You have a sensor that gives output as voltage as proportional to the light intensity. You need to switch off the light when the output of the sensor is above 5V. You need to switch on the light when the sensor output is below the 5V. (You can assume the V_H and V_L value)

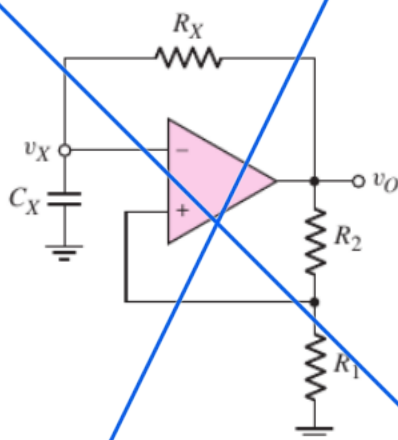
- Draw** the voltage transfer characteristics curve (V_{in} vs V_{out} **plot**). Clearly **label** the plot.
- Draw the circuit diagram that can be perfect for this specification.
- Find out the parameter value of the circuit.**

9. Suppose you want to design a street light controller. You have a sensor that gives output as voltage as proportional to the light intensity. You need to switch off the light when the output of the sensor is above 5V. You need to switch on the light when the sensor output is below the 5V. There is a noise source of 1V peak-peak. Your instructor has told you to use a schmitt trigger circuit to improve performance. (You can assume the V_H and V_L value)

- What V_{TH} and V_{TL} value can be used for this design to solve the problem of noise?
- Draw** the voltage transfer characteristics curve (V_{in} vs V_{out} **plot**) and clearly **label** the plot.
- Draw the circuit diagram that can be perfect for this specification
- Find out the parameter value of the circuit.

Square Wave Generator

1.



Here suppose $R_1 = 10k$, $R_2 = 20k$, $R_X = 1k$, $C_X = 1\text{ mF}$, $V_H = 10V$ and $V_L = -10V$

- Find the period and frequency of the square wave?

Comparison and Schmitt Trigger

1. a) Here, $V_- = 0$

$$V_+ = V_{REF} \times \frac{R_2}{R_1 + R_2} + V_I \times \frac{R_1}{R_1 + R_2}$$

$$= 2 \times \frac{10}{10 + 5} + V_I \times \frac{5}{5 + 10}$$

$$V_+ = \frac{4}{3} + \frac{V_I}{3}$$

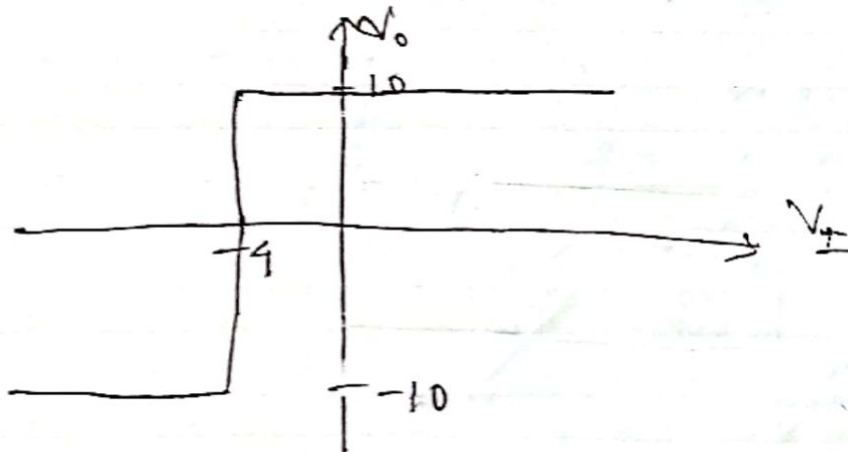
We know, $V_0 = V_H = 10V$

$$\text{i.e., } V_I > -\frac{R_2}{R_1} V_{REF}$$

$$\Rightarrow V_I > -\frac{10}{5} \times 2$$

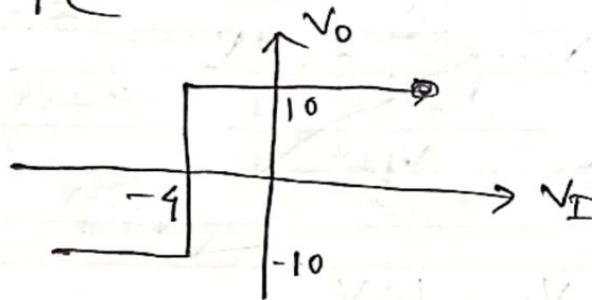
$$\Rightarrow V_I > -4 \longrightarrow V_0 = 10V$$

$$\text{So, } V_I < -4 \longrightarrow V_0 = -10V$$



1.6) This is a non-inverting comparator as the input is applied to the positive terminal.

VTC



From the graph if V_I is high V_O is also high
if V_I is low V_O is low

This is the characteristics of non-inverting comparator

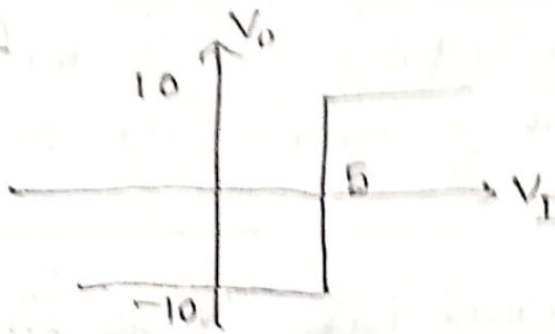


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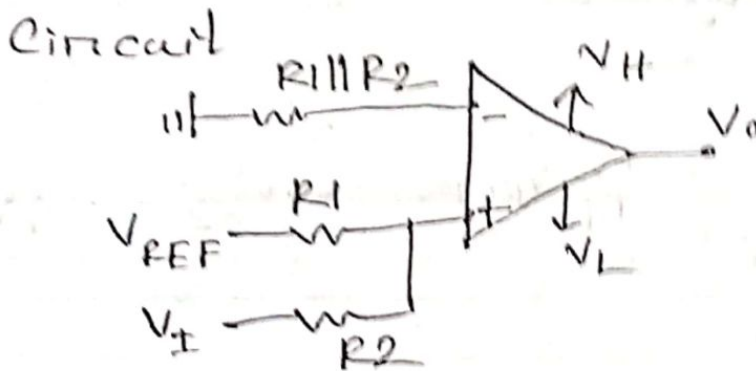
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2. Φ



This is a non-inverting comparator with V_{ref}



Now, From the VTC, $V_H = +10V$
 $V_L = -10V$

$$V_{Threshold} = 5 = -\frac{R_2}{R_1} V_{REF}$$

Suppose, $R_1 = R_2 = 10k\Omega$

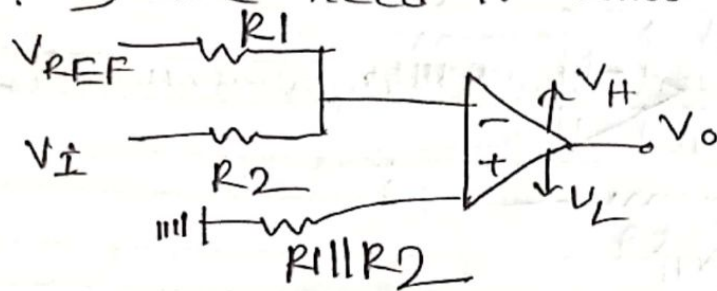
$$\therefore 5 = -\frac{10}{10} V_{REF}$$

$$\therefore V_{REF} = -5V$$

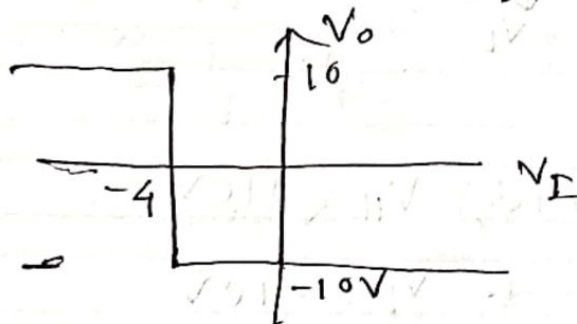
$$\therefore V_H = +10V, V_L = -10V, R_1 = R_2 = 10k\Omega, V_{REF} = -5V$$

3. a) This is an inverting comparator with V_{REF} as input is connected to negative terminal.

3. b) We need to know the VTC first,

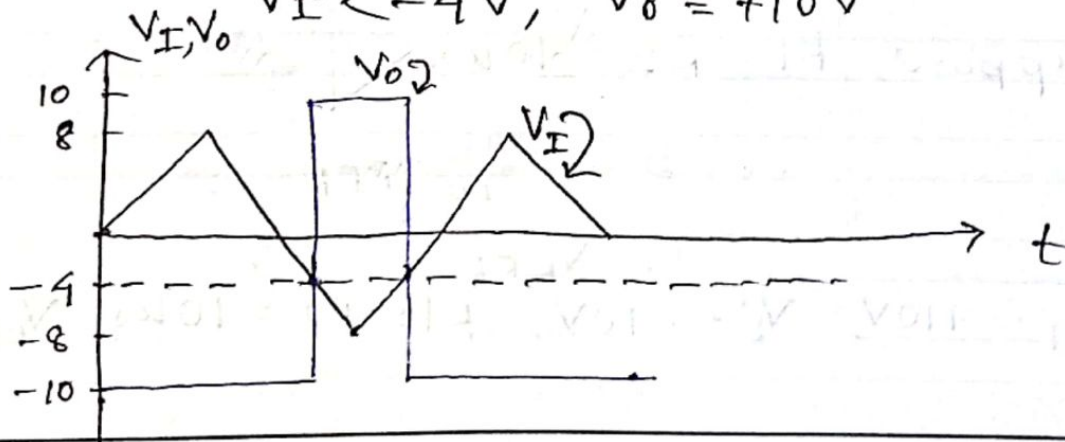


$$V_{\text{threshold}} = -\frac{R_2}{R_1} V_{REF} = -\frac{10}{5} \times 2 = -4V$$



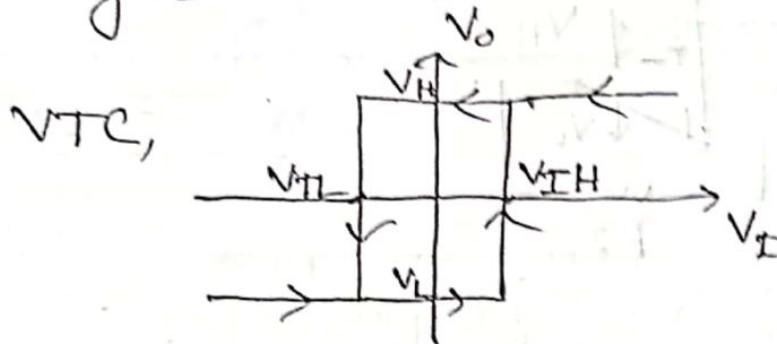
if, $V_I > -4V$, $V_O = -10V$

$V_I < -4V$, $V_O = +10V$

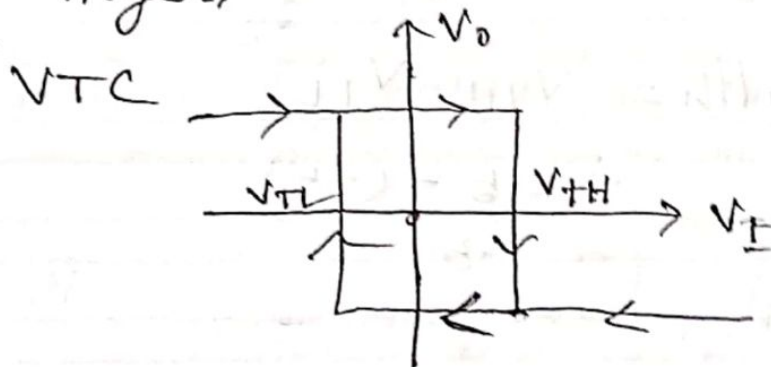


Output will be a square wave.

4.2) First circuit is non-inverting Schmitt trigger.



2nd circuit is inverting Schmitt trigger,



Their input output characteristics is inverse.



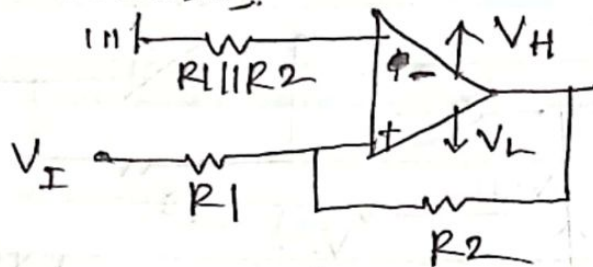
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b) Given, $R_1 = 5k$, $R_2 = 10k$, $V_{REF} = 2V$

$$V_H = 10V, \quad V_L = -5V$$

For circuit 1,



$$V_{TH} = -\frac{R_1}{R_2} V_L = -\frac{5}{10} \times -5 = 2.5V$$

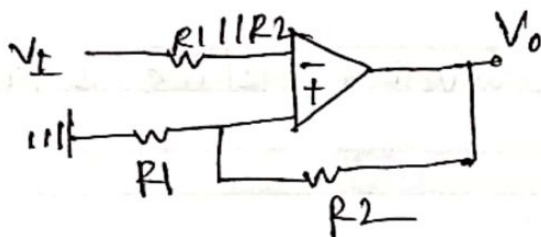
$$V_{TL} = -\frac{R_1}{R_2} V_H = -\frac{5}{10} \times 10 = -5V$$

$$\text{Hysteresis width} = V_{TH} - V_{TL}$$

$$= 2.5 - (-5)$$

$$= 7.5$$

For circuit 2,



$$V_{TH} = V_H \times \frac{R_1}{R_1 + R_2}$$

$$= 10 \times \frac{5}{5 + 10}$$

$$= 3.33 \text{ V}$$

$$V_{TL} = V_L \times \frac{R_1}{R_1 + R_2}$$

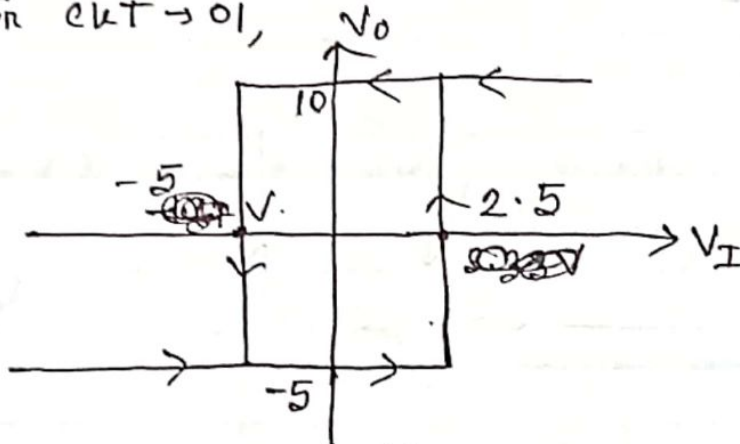
$$= -10 \times \frac{5}{5 + 10}$$

$$= -3.33 \text{ V}$$

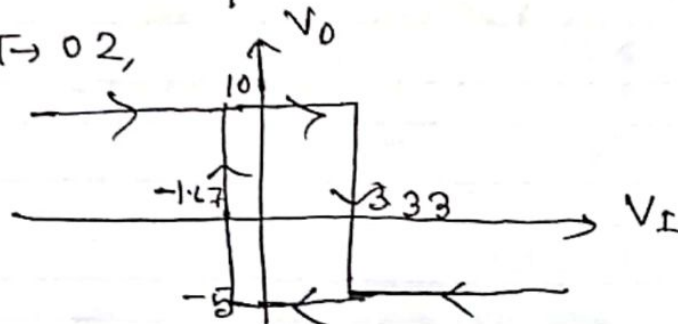
$$\text{Hysteresis width} = V_{TH} - V_{TL}$$

$$= 3.33 - (-3.33) = 6.66 \text{ V}$$

e) For $e_{UT} \rightarrow 01$,



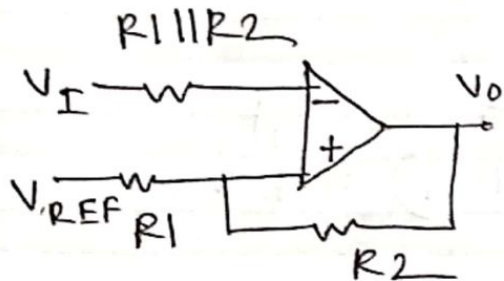
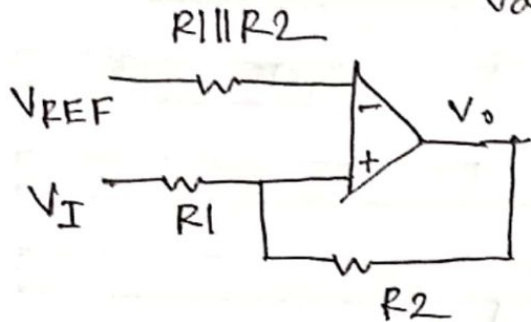
For $e_{UT} \rightarrow 02$,



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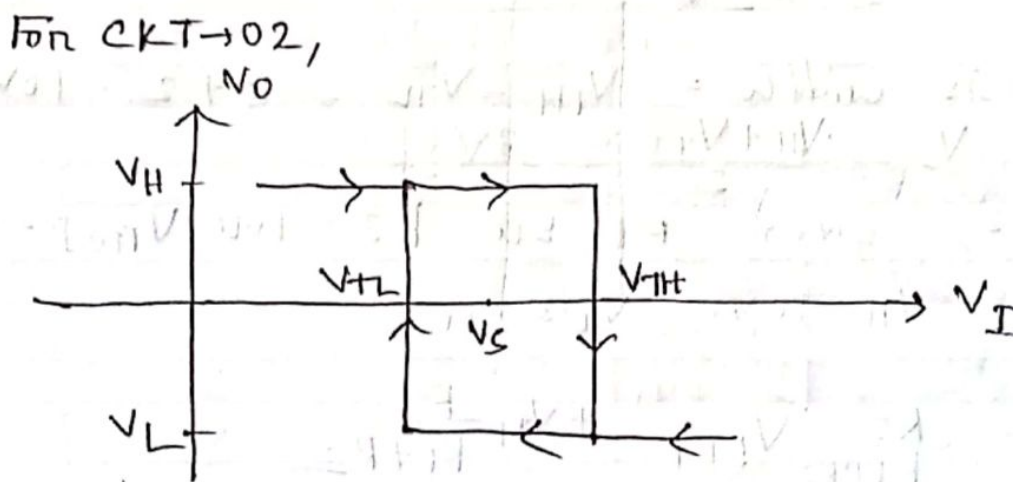
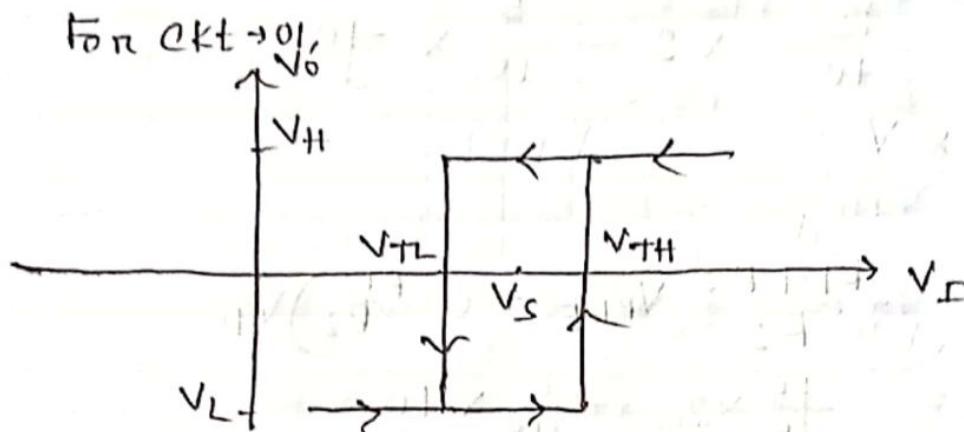
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d) If we can add V_{ref} , center voltage will shift. In another way we can use different value V_H and V_L .



5. a) Circuit 1, is the non-inverting schmitt trigger circuit with reference voltage

Circuit 2, is the inverting schmitt trigger circuit with reference voltage.



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5.6) For CKT1, given, $R_1 = 5k$, $R_2 = 10k$, $V_{REF} = 0V$
 $V_H = 10V$, $V_L = -10V$

$$V_{TH} = \frac{R_1 + R_2}{R_2} V_{REF} + \left(-\frac{R_1}{R_2} \right) V_L$$

$$= \frac{15}{10} \times 2 - \frac{5}{10} \times -10$$

$$= 8V$$

$$V_{TL} = \frac{R_1 + R_2}{R_2} V_{REF} + \left(-\frac{R_1}{R_2} \right) V_H$$

$$= \frac{15}{10} \times 2 - \frac{5}{10} \times 10$$

$$= -2V$$

Hysteresis width = $V_{TH} - V_{TL} = 8 + 2 = 10V$

$$V_S = \frac{V_{TH} + V_{TL}}{2} = 3V$$

For CKT2, given, $R_1 = 5k$, $R_2 = 10k$, $V_{REF} = 2V$
 $V_H = 10V$, $V_L = -10V$

$$V_{TH} = \frac{R_2}{R_1 + R_2} V_{REF} + V_H \frac{R_1}{R_1 + R_2}$$

$$= \frac{10}{15} \times 2 + 10 \times \frac{5}{15}$$

$$= 6.33V$$

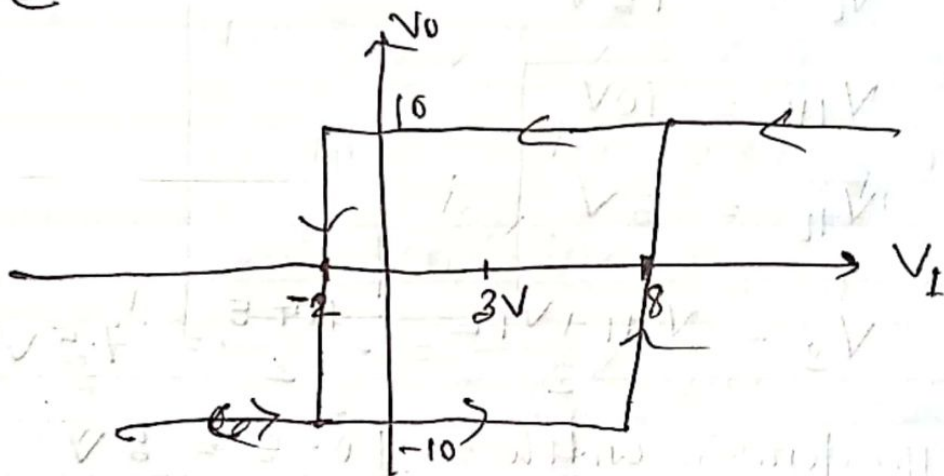
$$V_{TL} = \frac{R_2}{R_1 + R_2} V_{REF} + V_L \frac{R_1}{R_1 + R_2} = -3.66V$$

$$\text{Hysteresis width} = 6.33 + 3.66$$

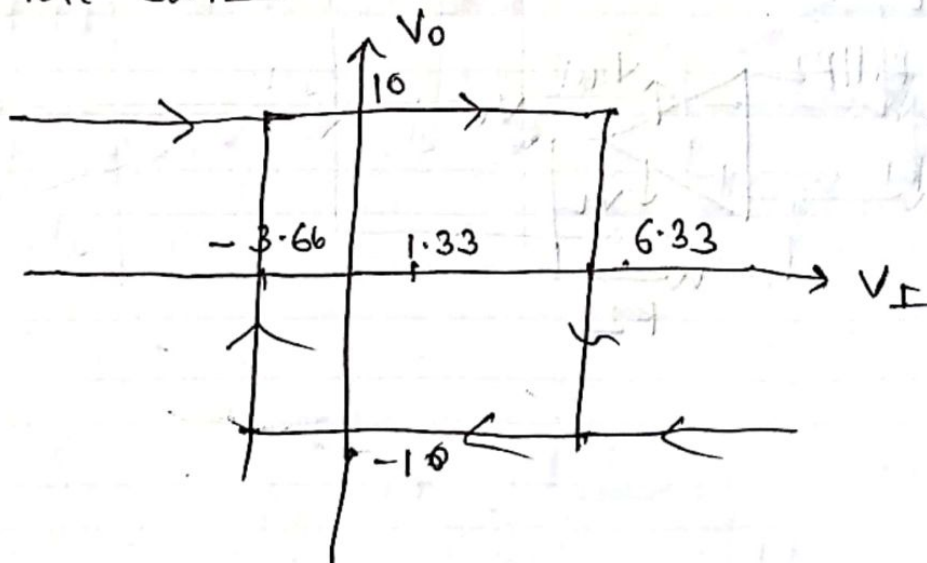
$$= 10 \text{ V}$$

$$V_S = \frac{6.33 + (-3.66)}{2} = 1.33 \text{ V}$$

5. a) For CKT 1,



For CKT 2



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6. a) This is a non-inverting schmitt trigger circuit

6. b) From the graph,

$$V_H = +15V$$

$$V_L = -15V$$

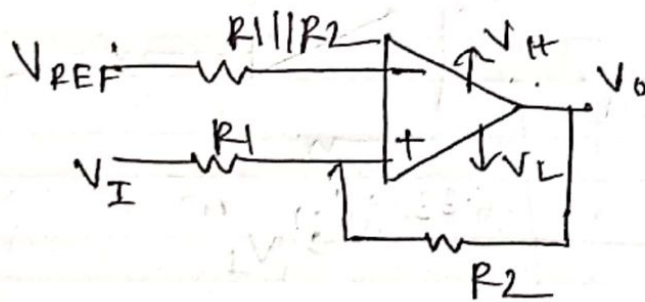
$$V_{TH} = 10V$$

$$V_{TL} = 5V$$

$$V_S = \frac{V_{TH} + V_{TL}}{2} = \frac{10 + 5}{2} = 7.5V$$

$$\text{Hysteresis width} = 10 - 5 = 5V$$

6. c) Circuit



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$V_H = +15V$ and $V_L = -15V$, These two are from graph.

$$V_{TH} = \frac{R_1 + R_2}{R_2} V_{REF} + \left(-\frac{R_1}{R_2}\right) V_L$$

$$V_{TL} = \frac{R_1 + R_2}{R_2} V_{REF} + \left(-\frac{R_1}{R_2}\right) V_H$$

$$V_S = \frac{V_{TH} + V_{TL}}{2}$$

$$\therefore V_S = \frac{R_1 + R_2}{2R_2} V_{REF} \rightarrow (1)$$

$$V_{Hysteresis} = V_{TH} - V_{TL} = \frac{R_1}{R_2} (V_H - V_L) \rightarrow (2)$$

From graph, $V_S = \frac{5+10}{2} = 7.5$

$$V_{Hysteresis} = 10 - 5 = 5$$

Suppose, $R_1 = 10k\Omega$,

From, (1), $5 = \frac{10k}{R_2} (15+15)$

$$R_2 = \cancel{40k}, 60k$$

From, (2), $5 = \frac{10}{40} \times V_{REF}$

$$\therefore V_{REF} = \cancel{5} 4.29V$$

7.a) This is inverting schmitt trigger with V_{REF} .

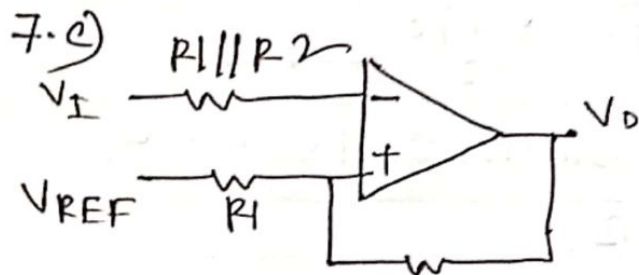
7.b $V_H = 15V,$
 $V_L = -15V$

$V_{TH} = +10V$

$V_{TL} = +5V$

$V_{Hysteresis} = V_{TH} - V_{TL} = 5V$

$V_S = \frac{V_{TH} + V_{TL}}{2} = \frac{10+5}{2} = 7.5$



$V_S = V_{REF} \times \frac{R_2}{R_1 + R_2} \rightarrow (1)$

$V_{Hysteresis} = (V_H - V_L) \frac{R_1}{R_1 + R_2} \rightarrow (11)$

We know, from graph
 $V_S = 7.5V$

$V_{Hysteresis} = 5V$

From, (11), $5 = (15 + 15) \frac{R_1}{R_1 + R_2}$
 $\frac{R_1}{R_1 + R_2} = \frac{1}{6} \Rightarrow \frac{R_1}{R_2} = \frac{1}{5}$

Now suppose, $R_1 = 1\text{ k}\Omega$,

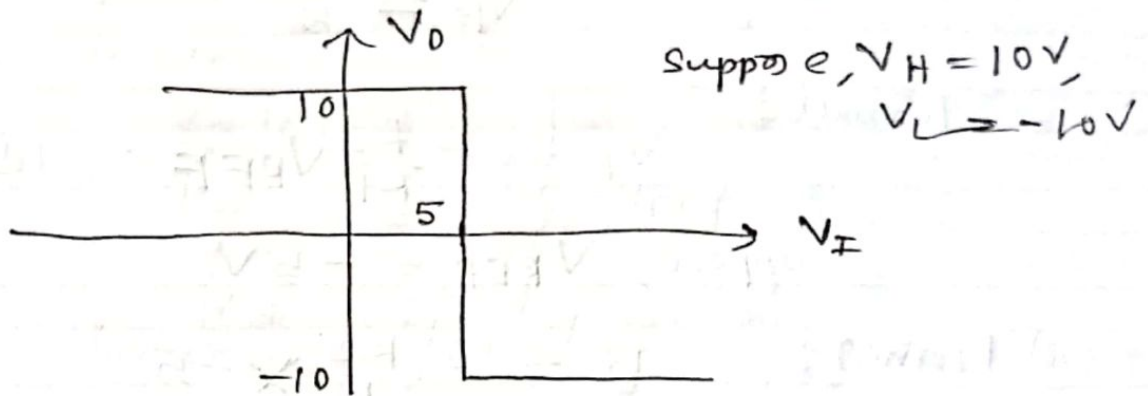
then, $\frac{1\text{ k}}{R_2} = \frac{1}{5}$

$$R_2 = 5\text{ k}\Omega$$

From ①, $7.5 = V_{REF} \times \frac{5}{1+5}$

$$\Rightarrow V_{REF} = +6\text{ V}$$

8. a)



8. b) There is single transition point, and input output is inverse.

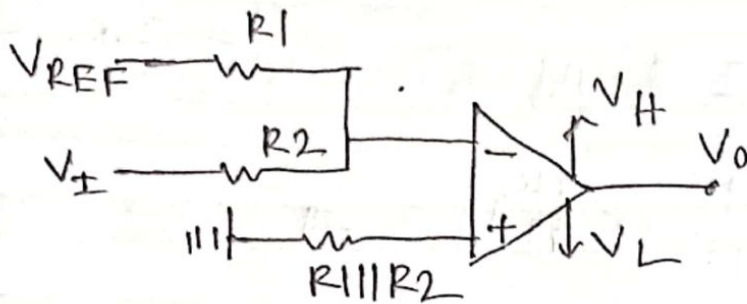
Inverting comparator with V_{REF} can be used



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8.c) From 8(a), $V_H = +10V$

$$V_L = -10V$$

$$V_t = 5$$

Formula,

$$V_t = -\frac{R_2}{R_1} V_{REF} \quad \text{--- (1)}$$

Suppose, $V_{REF} = -5V$,

$$\text{From (1),} \quad 5 = -\frac{R_2}{R_1} \times -5$$

$$\frac{R_2}{R_1} = 1$$

$$\text{If, } R_1 = 10k$$

$$R_2 = 10k$$

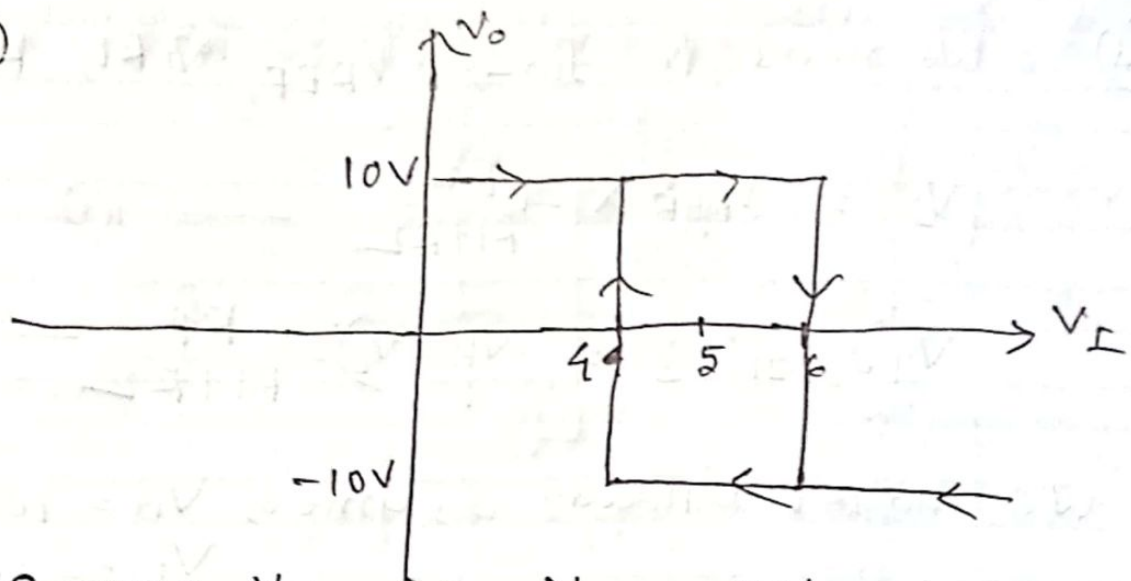


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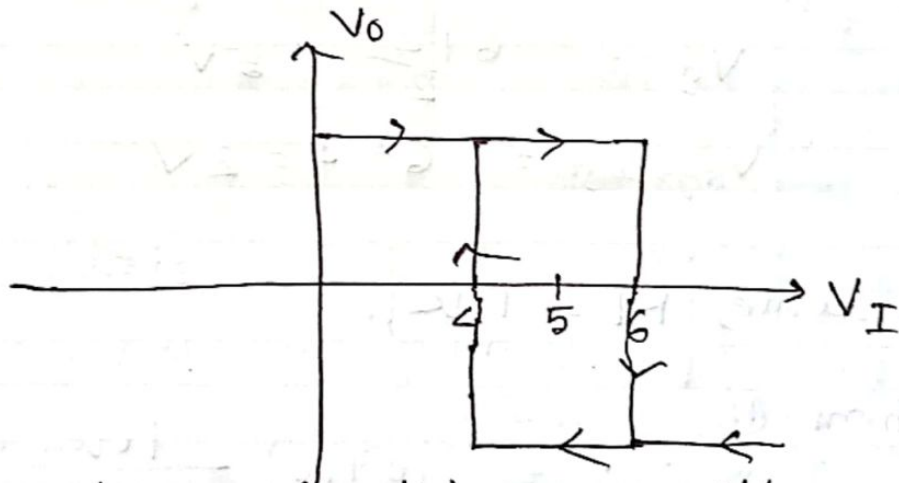
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9. a)

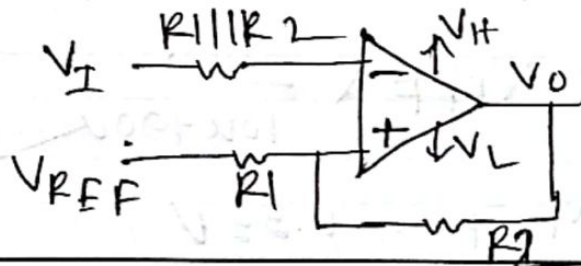


Suppose, $V_H = +10V$, $V_L = -10V$,
 $V_{THL} = 5 - 1 = 4V$,
 $V_{TH} = 5 + 1 = 6V$

9. b)



9. c) Inverting Schmitt trigger with V_{REF} can be used



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9.d) We need to find, V_{REF} , R_1 , R_2 .

$$V_S = V_{REF} \times \frac{R_2}{R_1 + R_2} \quad \text{--- (i)}$$

$$V_{Hysteresis} = (V_H - V_L) \frac{R_1}{R_1 + R_2} \quad \text{--- (ii)}$$

We have already assumed, $V_H = 10V$,
 $V_L = -10V$

From Equation,

$$V_{TH} = 5 + 1 = 6V$$

$$V_{TL} = 5 - 1 = 4V$$

$$V_S = \frac{6 + 4}{2} = 5V$$

$$V_{Hysteresis} = 6 - 4 = 2V$$

Now assume, $R_1 = 10k$,

From (ii),

$$2 = (10 + 10) \frac{10k}{10k + R_2}$$

$$R_2 = 90k$$

$$\text{From (i), } 5 = V_{REF} \times \frac{90k}{10k + 90k}$$

$$V_{REF} = 5.55V$$



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